

LINKEDIN JOB ANALYTICS DASHBOARD

GROUP 2 - DATA 201

ARYAN MITTAL - DATA ENGINEER

JASKIRAT SINGH - DATA ENGINEER

JAYANTH PASUPULATI - FULL STACK ENGINEER

PROBLEM STATEMENT

- Job seekers face information overload: 100,000+ postings with no easy way to compare salaries, skills, or industries
- There is no single tool that answers: "Which skills pay the most? Which industries are hiring? Does remote work pay more?"
- Raw LinkedIn data exists but is scattered across 11 separate CSV files – impossible to analyze without a database
- **GOAL:** Build a queryable MySQL database + interactive dashboard that turns raw LinkedIn postings into actionable job market insights

Job seekers cannot easily find patterns in salary, skills, and hiring demand across thousands of LinkedIn postings

Success Criteria

- A working dashboard that answers 12 distinct analytical questions from a single database `linkedin\_jobs`
- Any user can explore the job market in under 60 seconds without writing SQL or any technical acumen

DATASET

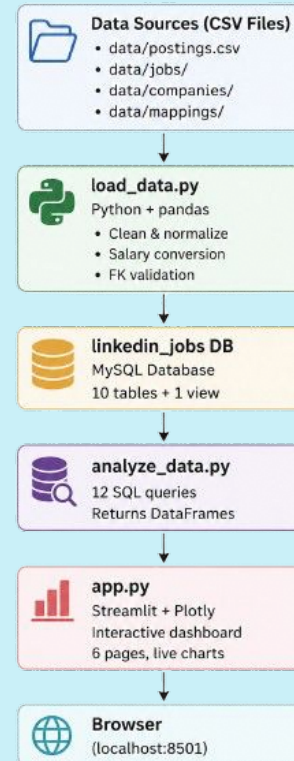
- Publicly Available LinkedIn Job Postings Dataset From Kaggle
 - Stored In CSV Files (data/)
- Covers Postings Listed Between 2023 - 2024
- **NOTE**
 - Only ~33% of postings have salary data (40,785 of 123,849)
 - Salary normalization: all pay periods (hourly/weekly/monthly) converted to annual USD equivalent
 - 0 Filled for NULL, Unknown for Experience Levels
- **LINK:**
<https://www.kaggle.com/datasets/arshkon/linkedin-job-postings>

METRIC	TOTAL COUNT
Job Postings	123,849
Companies	24,473
Industries	388
Unique Skills	35
Salary Records	40,785
Skill-To-Job Links	213,768

SYSTEM ARCHITECTURE

TECH STACK

- **FRONTEND/BACKEND:** PYTHON
- **DATA PROCESSING:** PANDAS
- **DATABASE:** MYSQL
- **QUERY:** SQLAlchemy +
mysql-connector-python
- **WEB FRAMEWORK:** Streamlit
- **CHARTS:** PLOTLY EXPRESS + PLOTLY
GRAPH OBJECT



DATABASE SCHEMA

- 10 TABLES
 - INDUSTRIES
 - SKILLS
 - COMPANIES
 - COMPANY_INDUSTRIES
 - EMPLOYEE_COUNTS
 - POSTINGS
 - SALARIES
 - JOB_SKILLS
 - JOB_INDUSTRIES
 - BENEFITS
- 1 VIEW
 - ANALYTICS: denormalized JOIN of postings + companies + salaries + industries + skills -> used for fast dashboard queries without repeating JOINS
- RELATIONSHIPS
 - POSTINGS -> COMPANIES
 - many-to-one
 - POSTINGS -> SALARIES
 - one-to-one
 - POSTINGS <-> SKILLS
 - many-to-many
 - POSTINGS <-> INDUSTRIES
 - many-to-many
 - COMPANIES <-> INDUSTRIES
 - many-to-many

IMPLEMENTATION

ETL PIPELINE (load_data.py)

- EXTRACT [Read all 11 CSV files]
- TRANSFORM [Clean nulls, deduplicate, normalize salaries to annual USD, validate FK references]
- LOAD [Insert into MySQL in dependency order (dimensions → facts → junctions)]; End up with linkedin_data.db file

ANALYTICS (analyze_data.py)

- 12 SQL query functions, each answering one business question
- Filter Outliers
- Returns pandas DataFrames [Clean interface between the database and the dashboard]

DASHBOARD (app.py)

- Renders Using Streamlit and Plotly
- 6 Pages
 - OVERVIEW
 - SKILLS
 - INDUSTRIES
 - COMPANIES
 - SALARY
 - JOB SEARCH
- Live search page dynamically queries the database on user input [No Pre-Loaded Data]

VISUALIZATION DEMO

CONCLUSION

KEY FINDINGS

- Remote jobs pay \$27,663/yr more than on-site [\$140,811 vs \$113,148]
- Executive roles earn 2.7× more than entry-level [\$221,475 vs \$81,289]
- Most in-demand skill ≠ highest paying – IT leads demand, Product Management leads salary [\$179K]

KEY LESSONS

- Real-World Data is Messy
- Importance of Relational Databases and Dashboards
- How SQL and Python Support Each Other

FUTURE WORK

- Make The Existing App Deployable
- Incorporate Recent Data
- Build NLP Features
- Geographical Analysis
- Role Specific Views

QUESTIONS?